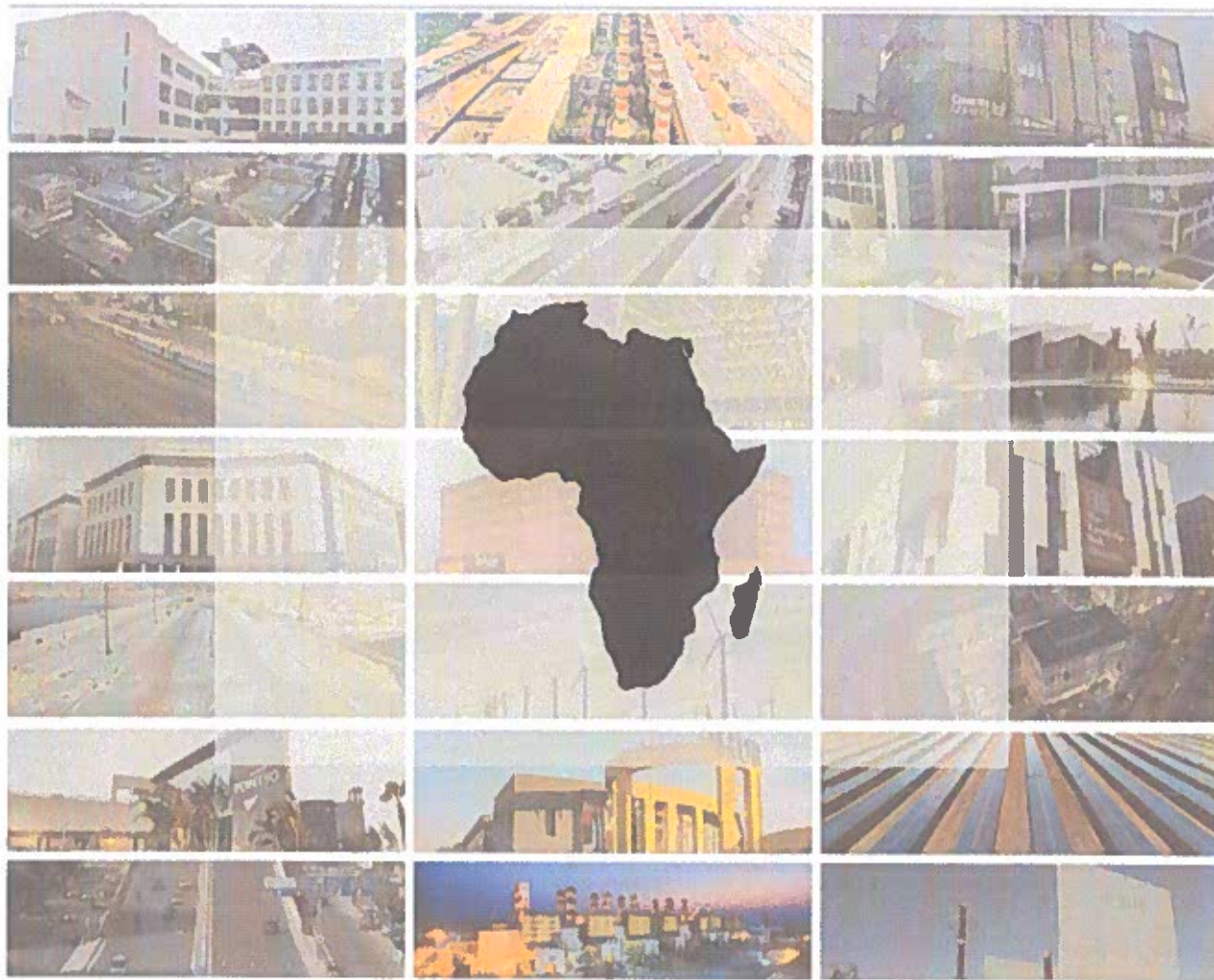




Formwork Process P-FW-01

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May, 2023

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1. OBJECTIVES

The objective of this process is to utilize formwork materials in the best optimized way in all Rowad's projects, and to level up the construction quality, by using the most effective engineering methods and projects management practices.

2. STAKEHOLDER

- Formwork department
- Project Team
- Warehouse
- Purchasing
- Tender
- Operation
- Logistics
- Quality Control
- COO

3. LIABLE STAKEHOLDER

- Formwork
- Project Team
- PM

4. TERMS AND DEFINITION

- **FW:** Formwork
- **RFP:** Request for Proposal
- **PM:** Project manager
- **CM:** Construction manager
- **TOM:** Technical office manager
- **FWCS:** Formwork Comparison sheet
- **SOR:** Service order request
- **PO:** Purchasing order
- **PR:** Purchasing request
- **RO:** Renting order
- **MTO:** Material Take-off
- **DC:** Document Controller
- **QC:** Quality Control

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- **IIR:** Internal Inspection Request
- **RFQ:** Request for Quotation
- **COO:** Chief operation officer
- **HoD:** Head of Department
- **HSE:** Health, Safety & Environment
- **SOG:** Slab on Grade

5. RELATED DOCUMENTED INFORMATION

- FW Overall Project Sheet, F-P-FW-01-01
- FW Balance sheet, F-P-FW-01-02
- Material Take-off, F-P-FW-01-03
- FW Calculation Note, F-P-FW-01-04
- FW Overall Rotation Plan, F-P-FW-01-05
- FW Management Plan, F-P-FW-01-06
- Overall Budget Study, F-P-FW-01-07
- FW Project Rotation Plan, F-P-FW-01-08
- FW Purchasing Report, F-P-FW-01-09
- FW Purchasing Lookahead, F-P-FW-01-10
- FW Evaluation Report, F-P-FW-01-11
- Service Order Request, F-P-FW-01-12
- FW Transfer & Receiving form, F-P-FW-01-13
- Service Order Request Log, F-P-FW-01-14
- FW Issue Form, F-P-FW-01-15
- FW Installation Checklist, F-P-FW-01-16
- FW Cost Report, F-P-FW-01-17
- FW Monthly Voucher, F-P-FW-01-18
- FW Maintenance Voucher, F-P-FW-01-19
- FW Evaluation Form, F-P-FW-01-20
- Project FW Final Report, F-P-FW-01-21
- FW Request for Quotation, F-P-FW-01-22
- FW Transfer request, F-P-FW-01-23
- Internal Inspection Request (IIR) – Falsework & Formwork, F-P-FW-01-24
- FW Material Inspection Request, F-P-FW-01-25
- Consumable Comparison sheet, F-P-PU-01-05

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- MOM, F-P-QD-06-02
- Internal MIR, F-P-QD-07-01
- Attendance Sheet, F-PQP-01-15
- FW SC Appendix 1
- PR (ERP)
- PO (ERP)

6. PROCESS DETAILS

6.1 PROCESS INPUT

- Project's awarding notification.
- IFC drawings.
- Project Baseline.
- Project's budget sheet.
- FW technical data sheets.
- Approved FW suppliers.

6.2 Process Operations

Sr.	Process Cycle Description	Responsibility	Form
Tendering			
1	As per Tender request, FW shall send RFP via e-mail to all concerned suppliers, and set formwork's cost, and sequence of work as per provided project duration provided by Tender engineer.	Formwork	
Preparation "After Project Awarded"			
2	FW Design Team shall study the whole project in general with extracting main design concept of FW, and the expected required quantities of material.	FW Design Team	FW Overall Project Sheet F-P-FW-01-01
3	FW shall conduct a meeting with PM/CM within two weeks from the tender handover meeting to discuss the following: - Project commencement date - Final sequence of work and FW rotation - Budget	Formwork PM	Minutes of Meeting F-P-QD-06-02

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	- Utilized systems that will be used - Material delivery dates - Any special concerns		
4	FW shall issue overall rotation plan for FW systems in all Rowad projects with 4 months lookahead, to be updated on monthly basis and submit it to the COO.	FW Manager	FW Overall Rotation Plan. F-P-FW-01-05
5	FW material control section head shall check for material availability in FW main yard and releasing projects through FW balance sheet and overall FW rotation plan. FW design team shall design according to material availability, and submit the following to the project team (Document Controller, TOM) after the approval of FW manager for client's approval: - Design Drawings - Calculation Notes - MTO	FW Material Control Section head FW Design Team	FW Balance Sheet F-P-FW-01-02 FW Overall Rotation Plan F-P-FW-01-05 FW Calculation Note F-P-FW-01-04 MTO F-P-FW-01-03
6	FW shall issue FW management plan within one month from Tender handover meeting and update quarterly for each project including but not limited to: - Project Summary - IFC drawings reference number - Rowad scope and S/C scope - FW rotation plan for project according to project time schedule provided from project control department - FW overall budget study - Formwork breakdown structure - Responsibility matrix. - Material management plan: • Systems that will be used • Items that will be purchased/rented/supplied internally • Delivery plan	FW Manager PM	FW Management Plan F-P-FW-01-06 FW Overall Budget Study F-P-FW-01-07 FW Project Rotation Plan F-P-FW-01-08 MTO F-P-FW-01-03

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	- FW design drawings, calculation notes, and MTO		
7	FW manager and PM shall approve FW management plan and submit to COO, Tender & Projects Control.	FW Manager	FW Management Plan F-P-FW-01-06
FW Purchasing Process			
8	FW shall check the available material and prepare a list for the required items to be requested by the project.	FW	MTO F-P-FW-01-03
9	Purchasing Request (PR) shall be issued by project's DC via ERP according to the received MTO. Refer to Purchasing process (P-PU-01)	Project's DC	PR (ERP)
10	FW Purchasing Officer provide quotations from at least three suppliers for the requested items and issue FWCS. Refer to: Vendor Selection and Bidding Process (F-P-VMO-01) Evaluation of External Providers (P-PR-02)	FW Purchasing Officer	FW Request for Quotation F-P-FW-01-22 Formwork Comparison Sheet F-P-FW-01-27
11	FW Purchasing Officer shall submit FWCS to FW Manager for his approval and supplier selection. Refer to CS Signature Matrix.	FW Manager	Formwork Comparison Sheet F-P-FW-01-27 CS Signature Matrix
12	FW Purchasing Officer shall issue PO/RO within 5 working days after FWCS is approved.	FW Purchasing Officer	PO (ERP) RO (ERP)
13	FW shall issue FW purchasing report quarterly includes but not limited to: - Purchased value - Saving value To be approved by FW Manager and submitted to COO.	FW Purchasing Officer FW Manager	FW Purchasing Report F-P-FW-01-09

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14	FW shall issue 3 months purchasing forecast on quarterly basis for all formwork systems according to the overall rotation plan and submit to COO.	FW Manager	FW Purchasing Lookahead F-P-FW-01-10
Execution Process			
15	FW DC shall send SOR to FW logistics coordinator includes the items that required to be supplied from another project or FW yard to the concerned project according to site request and FW project management plan.	FW DC	Service Order Request F-P-FW-01-12
16	FW Logistics Coordinator shall coordinate with Logistics department in the following cases: - Transfer items mentioned in SOR/PO to the concerned project on the agreed delivery dates. - Transfer FW items from releasing projects to FW yard. FW Logistics Coordinator shall update SOR log accordingly and submit it to FW manager on weekly basis.	FW Logistics Logistics	FW Transfer Request F-P-FW-01-23 SOR log F-P-FW-01-14
17	FW maintenance section shall inspect all received items in FW main yard to classify it into the following: - Items need maintenance. - Damaged items. - Items ready to be reused.	FW Maintenance Section Head	FW Yard Evaluation Report F-P-FW-01-11
18	FW maintenance shall repair any defect items and re-inspect, then add it to FW balance sheet and update it on weekly basis to be submitted to FW Manager.	FW Maintenance section Head	Internal MIR F-P-QD-07-01 FW Balance Sheet F-P-FW-01-02
19	FW Project Material Control receives the transfer items from suppliers/FW yard/other projects to FW project yard and proceeds with delivering to construction as per PM's approval.	FW Material Control FW Engineer PM	FW Transfer and Receiving Form F-P-FW-01-13 FW Issue Form F-P-FW-01-15

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20	During project execution, Site Engineer shall raise IIR minimum 3 days prior to pouring (except for SOG & plain concrete) to QC engineer & Formwork engineer to follow up & inspect the formwork which will be used for concrete elements, and after QC approval, QC shall inform formwork engineer for the final inspection, IIR to be approved by site, QC and FW engineer. A copy of the above IIR to be sent to HSE.	Site Engineer QC Engineer FW Engineer	FW Installation Checklist F-P-FW-01-16 Internal Inspection Request (Falsework & Formwork) F-P-FW-01-24
21	For subcontractors not using Rowad's FW Material the following shall be implemented: - FW standard, specifications, and minimum requirements to be developed by FW dept. and incorporated in the subcontractor's agreement. Accordingly, PM shall inform the S/C officially regarding Rowad standards. - After construction team's inspection and approval, shall inform QC and Formwork Engineer via FW Material Checklist upon receiving S/C (FW Material) for every delivery on site. - QC DC shall maintain FW Material Checklist log. - S/C shall submit the FW design drawings to be verified by FW engineer.	PM QC FW QC DC	Formwork Material Inspection Request F-P-FW-01-25 FW Material Checklist Log F-P-FW-01-26 FW SC Appendix 1
22	FW Material Control Section Head shall submit monthly vouchers of all projects according to each project FW balance sheet to FW manager for approval and submit to each PM for approval and TOM for information. FW shall update daily site report for all projects, and forward it to TOM, project control engineer.	FW Material Control Section Head PM	FW Balance Sheet F-P-FW-01-02 FW Monthly Voucher F-P-FW-01-18
23	FW Material Control Section Head shall submit maintenance voucher quarterly of all projects to FW manager for approval and submit to each PM for approval and TOM for information.	FW Manager FW Maintenance Head PM	FW Maintenance Voucher F-P-FW-01-19

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24	FW engineer shall update Project's rotation plan and FW Cost Report according to any changing orders, then submit it to PM.	FW Engineer PM	FW Project Rotation Plan F-P-FW-01-08 FW Cost Report F-P-FW-01-17
25	FW Engineer shall evaluate any damaged timbers and plywood that exist in projects and submit FW Evaluation Form to the Warehouse. Refer to Material Control Process (P-FN-01)	FW Engineer Warehouse	FW Evaluation Form F-P-FW-01-20 FW Transfer Request F-P-FW-01-23
Project Closure			
26	FW Manager shall issue project final report at the end of each project includes but not limited to: - Project summary - Final cost report - Actual FW rotation plan - Purchased, rented, and internally supplied items - Lessons learned	FW Manager	

6.3 PROCESS OUTPUT

- FW management plan.
- FW Project Final report.
- Technical Submittals.
- Monthly and maintenance vouchers
- FW overall rotation plan.
- FW purchasing report and lookahead

7. PROCESS RISK ASSESMENT

Included in the risk register form No. F-P-QD-05-01 "Risk identification and assessment" maintained by the HoD, along with the action plan to address risks and opportunities".

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8. PROCESS CONTROL

Formwork Manager shall review and approve the following:

- FW management plan.
- Technical submittals to projects.
- Project's FW cost report.
- FW overall rotation plan.

9. PERFORMANCE MEASURES

- Actual dates for submittals, purchasing and delivery dates versus planned dates.
- Percentage of cost saving from approved budget if applicable.
- Annual Percentage of repaired materials from Total RME materials.

10. DOCUMENT CHANGE HISTORY

No.	Revision Level	Description of Changes	Effective Date
1	00	New Issue	08/02/2022
2	01	Points 20 & 21	21/05/2023

ROWAD
MODERN ENGINEERING